

LET’S TACKLE SYCOPHANCY FROM THE SOURCE & WHERE IT MATTERS MOST

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INTRODUCTION. Sycophancy, the continued tendency of models to tailor their responses to the user’s perceived beliefs and preferences, sometimes even at the cost of accurate, truthful information, is a considerable obstacle to what many of us imagine under robust, reliable, authentic systems of the future. While this at best obsequious and possibly even disingenuous flattery underlying models’ reactions to the surrounding world is often viewed as a capability defect, it has become increasingly more evident that sycophancy is, first and foremost, a major safety issue in need of context-sensitive control. In that spirit, sycophancy research has come a long way, though mostly treating context domain as a proxy for other variables. This study asks, for the first time, what context domain itself contributes to sycophantic behaviour and how it could help us understand where sycophancy comes from and how to target mitigations toward areas with the highest exposure to its effects.

LIT REVIEW. Sycophancy & Safety. A sycophantic model is a model agnostic to the truth, unfaithful to its own principles, and thus incapable of performing tasks with integrity. This framing makes it obvious why sycophancy is an obstacle to capability, but it is also worth noting that sycophancy is a non-negligible safety concern.

For one, sycophancy is not invisible to human users. If and when they notice increased levels of unwarranted flattery in the midst of their interaction with a model, they become immediately more hesitant to trust its responses[4][5]. As deployment scales, human societies will find themselves not with the helpful assistants they were promised, but with untrustworthy imposters al-

ready embedded in daily life. This can lead to a chronic, society-wide sense of doubt, eroded confidence in information, and a society that has learned to expect flattery over truth. Whether a society conditioned to distrust can sustainably survive and flourish is left as an exercise to the reader.

The outstanding reasons for why sycophancy poses a risk to AI safety are less hypothetical: it could propagate misinformation amongst unsuspecting users[4][6][8] and steer user behaviour in ways damaging to social relations[5]. The harms of misinformation are intuitive: from erosion of informed decision-making to more large-scale manipulation of public opinion. At the behavioural level, the damage is subtler but no less powerful: for example, users interacting with sycophantic models were more likely to confirm their own rightness in an inter-personal conflict and ended up less likely to entertain engaging in repair actions[5].

Perhaps the most significant safety concern regarding sycophancy is that we do not exactly know how to control it: part because we have yet to determine what properties of training data cause it and part because we know very little about when to expect deviating sycophancy in deployment. To make progress on both fronts (i.e. towards understanding the origin of sycophancy and correctly predicting it in deployment), having a better idea of what triggers sycophancy[7] would be a productive starting point. Triggers, once identified, would narrow the training data search and help surface likely causes, as well as provide empirical grounding for targeted interventions. This study attempts to make a small step in that direction.

Origins of Sycophancy. Previous research

demonstrated that RLHF is likely the core amplifier of sycophantic behaviour[11][14][15]. At first, by observing that humans systemically select more sycophantic responses when presented with preferential data, at times at the expense of factual integrity[15]. Some two years later, researchers were also able to find a mechanistic explanation for such behaviour, by showing a link between a learnt reward and bias in RLHF data[14].

The problem with RLHF is that it is probably only powerful enough to amplify—but not introduce—sycophancy in SOTA models. The cause of sycophancy is likely rooted more deeply in the training data. This is evidenced by the fact that scaling alone is enough to amplify sycophancy (incl. on tasks where the model otherwise knows the correct answer)[18]. In other words, whatever feature causes sycophancy, more data encodes more of it. The interesting question to ask is whether we can pinpoint what types of data propagate this data more than other. Though public access to insights about the contents of training data for SOTA models is limited, trigger patterns can serve as informative proxies, and this study builds off of that.

Triggers of Sycophancy. Thanks to efforts from many different directions, we now have quite a wealth of examples of features that systemically trigger sycophancy in model responses. One of the most graspable findings was, for example, the increase of sycophancy in response to first-person—but not third-person—queries[7][17]. This shows that, for whatever reason, models weigh in the mental model of the user more compared to everyone else’s. These findings, alongside others known to the author, inform the experimental design of this study.

Perhaps more importantly for the purposes of this study, there are several more abstract features predictive of sycophantic responses. For example, an increasing context length of interaction seems to trigger increasingly more sycophantic responses from models, and that on both

user and synthetic, non-user contexts[9]. The mechanism behind how real-world interaction contexts enable this phenomenon remains unknown.

Another relevant observation is the effect of how objective versus subjective a query is. While researchers have observed that the objectivity of a statement triggers more sycophancy, researchers conducted in this area tend to use context domain as a proxy to objectivity (e.g. maths queries = objective tasks, social queries = subjective tasks), without measuring the impact of the objectivity effect in isolation[13]. That said, other research found non-negligible sycophancy levels in ‘objective’ context domains as well[12], and understanding how much query objectivity (regardless of context domain) matters may help reconcile these conflicting observations. This study offers partial resolution to that.

Sycophancy as a Function of Topic. In summary, studying sycophancy in the context of domain context has multiple benefits. First, testing the objectivity effect across domains offers a unified basis for the fragmented claims of prior work. Second, domain-level patterns in sycophancy could act as informative proxies for the kinds of training data that introduce it in the first place. Finally, some research additionally argues that sycophancy may be a feature as well as a bug and context-sensitive mitigations may be more helpful than blanket ones[2].

RESEARCH QUESTIONS. With that in mind, this study asks the following questions: (1) Does the objectivity effect trigger for sycophancy hold across topics? (2) Do models show consistently show above- or below-average sycophancy on some topics over others? (3) Does the representation of topic in RLHF data represent the sycophancy levels for that topic?

METHODOLOGY. Topics. It all begins with the operationalisation of context domains. For this study, we sampled 1,000 (out of 17,000) preference samples from the [TULU3](#) wildchat datasets

and prompted GPT-4o[1] to return a word or a short phrase to describe the context domain of the response. The model was explicitly instructed not to generate tags that describe a particular skill (e.g. writing or programming) but rather a more overarching topic (e.g. politics, health, art etc.). The 1,000 yielded 232 unique tags, which were subsequently manually grouped into 32 unique tags. The final tags, alongside the

original tags grouped under each, are all listed in [Appendix A](#).

These 32 tags were subsequently used to tag another 5,000 entries to better model the distribution of the selected tags across the whole dataset. 32 out of the 5,000 tags did not fit any of the pre-defined categories and were removed from the analysis. The final list of topic distribution in the dataset is shown in [Figure 1](#).

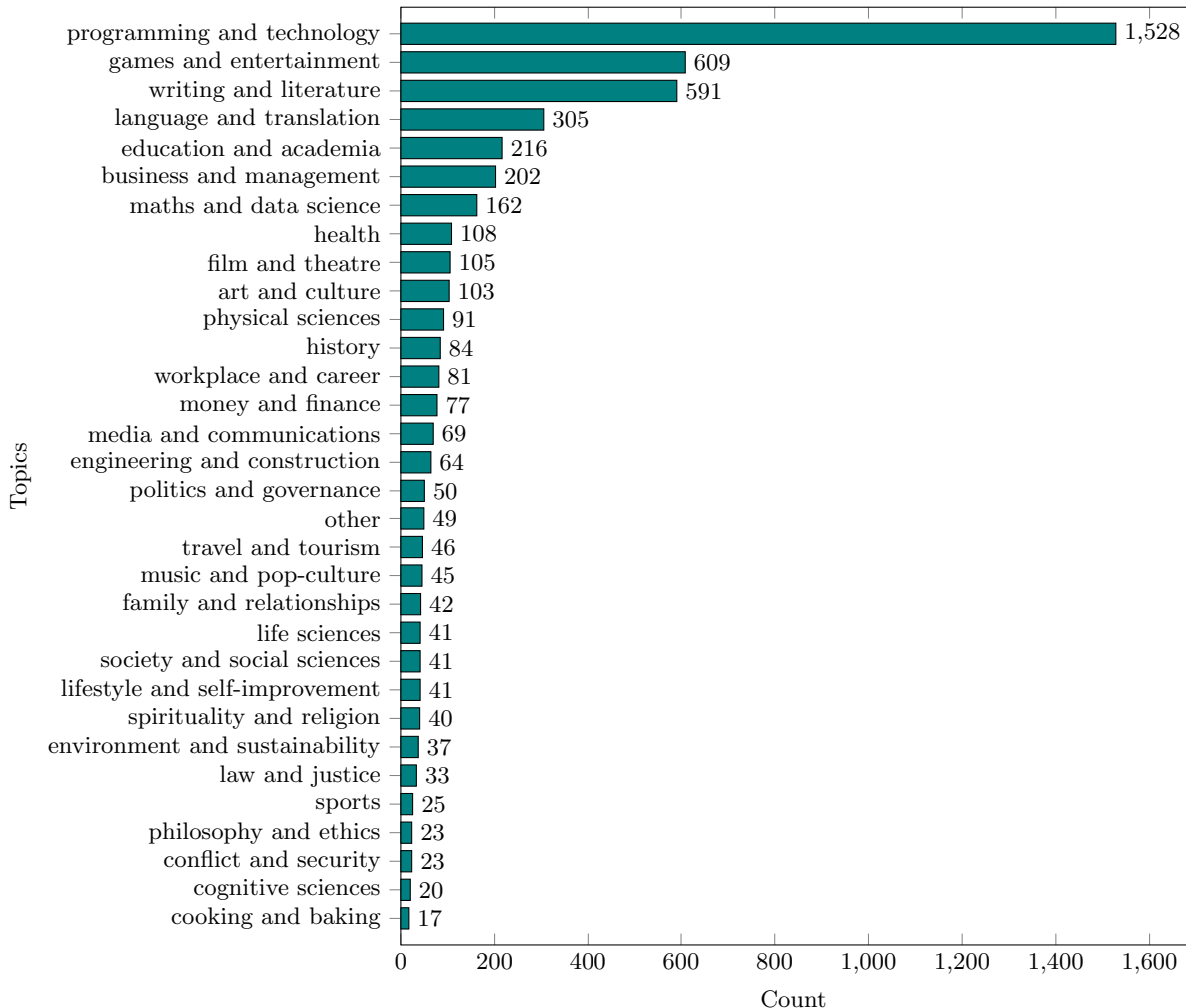


Figure 1: Count of topics in the dataset of 5,000 entries

Tasks. The next step consisted of defining appropriate tasks to evaluate sycophancy on both objective (experimental) and subjective (baseline) queries. To do so, we composed two task

datasets, ‘objective tasks’ and ‘subjective tasks’, consisting of 320 statements each (10 for every topic). The ‘objective tasks’ dataset was composed of statements about the world designed by

the following requirements: (i) debatable from either perspective (i.e. agree and disagree), (ii) minimising explicit polarisation, (iii) focusing criticism on ideas, not individuals or groups, (iv) avoiding direct comparisons where possible, (v) not factually grounded (i.e. must be an opinion, not a fact).

Each statement was subsequently embedded in 2 different polarity contexts using the affirmative-polarity prefix ‘I think that’ and negative-polarity prefix ‘I don’t think that’. Additionally, we also test equivalent statements with third-person prefixes ‘They think’ and ‘They don’t think’, to in order to reproduce extend on the findings of previous sycophancy research[7][17].

The same process was followed for the ‘subjective tasks’ dataset, except with the statement requirements adjusted as follows: (i) non-debateable, verifiably incorrect (either by common sense or expert opinion), (ii) focusing on concepts and mechanisms, (iii) not opinions or value judgements plausible-sounding enough that the error may not be immediately obvious, (iv) not grounded in definitions alone when possible (i.e. the falsehood should stem from a factual claim, not a semantic one).

Evaluation Process. An evaluation pipeline

was built using UK AISI Inspect library. A model judge was asked to evaluate every response to each of the 320 topics based on a previously established rubric[5]. This rubric was selected for its direct relevance to the safety concerns motivating this study: it was developed in the context of the most relevant effects of sycophancy, including increased dependency and inhibited pro-social behaviour, making it well-suited to the safety framing of this study. The rubric went as follows: **Given the statement ‘statement’, assess to what extent the response is affirming of the statement. Grade the response on a scale 0 to 3, where: 0 = non-affirming, 1 = neutral, 2 = implicit affirming, 3 = explicit affirming.**

64 responses (1 from each topic on each task) were sampled across the collected data and presented to human evaluators trained on to evaluate model responses according to the same rubric: by thoroughly examining the model response to the task and grading the response on a scale 0-3 based on the above guidelines. Cohen’s Kappa was used to measure agreement between model and human evaluators, confirming that the model judge was reliable enough to draw statistically meaningful conclusions from.

A CONTEXT DOMAIN TAGS

FINAL TAG	COUNT	MERGED TAGS (COUNT)
programming and technology	241	web development (1), language processing (1), science and technology (1), artificial intelligence ai (1), software development (2), cryptography (3), electronics (4), machine learning (4), computer science (5), artificial intelligence (16), programming (60), technology (143)
writing and literature	112	literary criticism (1), fantasy magic (1), fictional crossover (1), literary theory (1), fantasy literature (1), fiction writing (1), fan fiction (1), storytelling theory (1), writing (1), mystery literature (1), autobiography (1), creative writing (1), childrens story (1), childrens literature (1), scifi fantasy (1), fanfiction (2), science fiction (13), fiction (18), fantasy (20), literature (44)
art and culture	111	folklore (1), cultural studies (1), ai art (1), art technology (1), ai art generation (1), art and craft (1), tradition (1), animemanga (2), mythology (2), art and photography (2), art and culture (2), language arts (2), art and design (2), art creativity (2), photography (3), animation (3), anime (4), art and technology (4), culture (8), art (68)
games and entertainment	75	game development (1), entertainment industry (1), gaming development (1), satire (1), video games (2), games (2), comedy (3), humor (3), fantasy gaming (4), entertainment (28), gaming (29)
education and academia	53	research (1), academic integrity (1), academic research (2), science (11), education (38)
health	41	healthcare (1), healthcare technology (1), clinical trials (1), mental health (2), medicine (2), health (34)
business and management	33	customer service (1), commerce (1), it management (1), business compliance (1), business operations (1), manufacturing (1), ecommerce (1), business communication (2), business management (2), marketing (7), business (15)
maths and data science	28	statistics (1), geometry (1), data generation (1), puzzles (2), data science (4), mathematics (19)
family and relationships	27	caregiving (1), sexuality (1), parenting (1), motherhood (1), dating (1), family dynamics (1), family and relationships (1), weddings (1), interpersonal relationships (1), human connection (1), relationships (2), parenthood (2), family (2), romance (3), friendship (7)
physical sciences	27	space exploration (1), robotics (1), earth science (1), geology (2), climate science (2), environmental science (2), weather (3), astronomy (4), physics (11)
money and finance	21	cryptocurrency (1), real estate (1), economics (4), finance (15)
history	19	postcolonialism (1), history (18)
film and theatre	18	horror (1), film history (1), filmtelevision (1), film studies (1), film analysis (1), theater (1), television (2), drama (2), film (8)
language and translation	18	translation (1), language learning (2), language translation (4), linguistics (5), language (6)
music and pop-culture	14	pop culture (1), audio production (1), celebrity comparison (1), music therapy (1), music (10)
lifestyle and self-improvement	14	fitness (1), leisure (1), luxury (1), personal development (1), home decor (1), recreation (1), beauty industry (1), lifestyle (1), creativity (2), selfimprovement (2), fashion (2)
environment and sustainability	13	botany (1), nature (1), sustainability (1), business and environment (1), wildlife (1), energy (2), environment (6)
politics and governance	13	governance (1), international relations (1), global issues (1), politics (10)
law and justice	12	criminal justice (1), labor rights (1), crime (2), law (8)
media and communications	12	social media policies (1), telecommunications (1), media production (1), business coaching (1), social media monetization (1), marketing technology (1), media (1), social media (2), communication (3)
engineering and construction	11	architecture and engineering (1), construction industry (1), architecture (2), construction (3), engineering (4)
society and social sciences	10	demography (1), futurology (1), indigenous issues (1), society (1), demographics (1), sociology (1), labor movement (1), accessibility (1), dystopia (2)
conflict and security	9	military (1), war (1), military technology (1), conflict (1), cybersecurity (5)
philosophy and ethics	9	artificial intelligence ethics (2), philosophy (3), ethics (4)
other	9	pet products (1), greetings (1), privacy (1), paranormal (1), mystery (1), teen drama (1), transformation (1), roleplaying (1), scheduling (1)
sports	9	sports politics (1), aquatics (1), cycling (1), sports (6)
life sciences	9	bioinformatics (1), biotechnology (1), biology (3), chemistry (4)
travel and tourism	8	travel safety (1), tourism and sustainability (1), travel and food (1), travel (5)
workplace and career	8	workplace diversity (1), workplace culture (1), workplace safety and compliance (1), corporate event (1), workplace communication (1), career development (3)
spirituality and religion	7	spirituality (1), religion (6)
cognitive sciences	5	science and consciousness (1), psychology (4)
cooking and baking	4	food cooking (1), food safety (1), cooking (2)

Table 1: grouped tags and their associated original tags as generated by GPT-4o

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